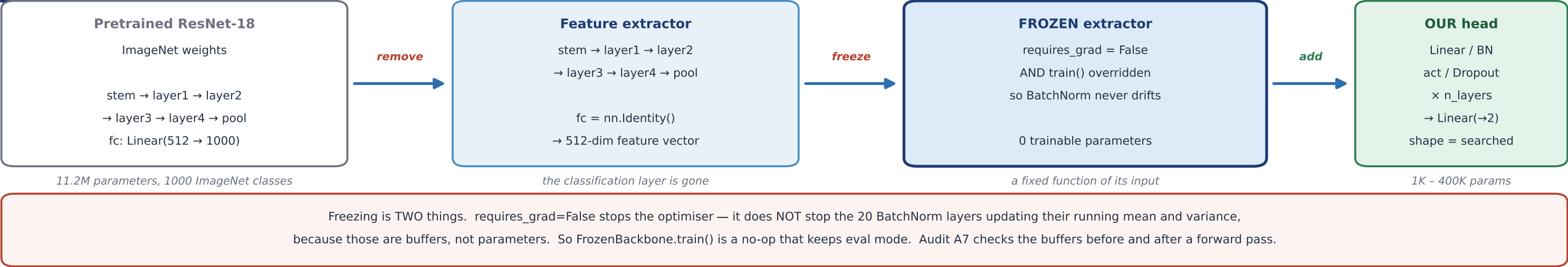


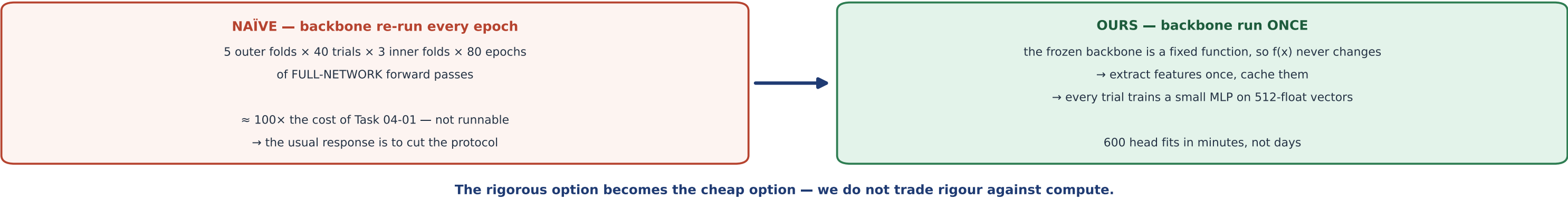
Task 10 · Frozen feature extractor + Optuna head search

Dr. Sattar: use only the feature-extraction layers, remove the classification layers, add your own, and tune them automatically with Optuna

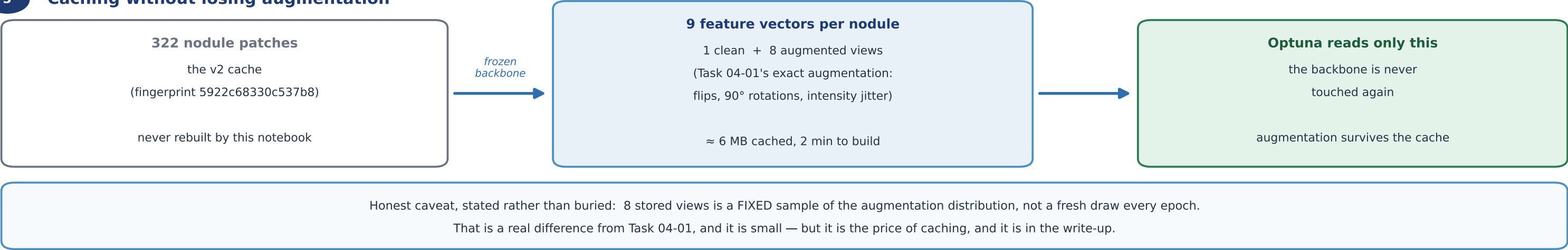
1 The surgery — what we remove, what we freeze, what we add



2 The move that makes an honest search affordable



3 Caching without losing augmentation

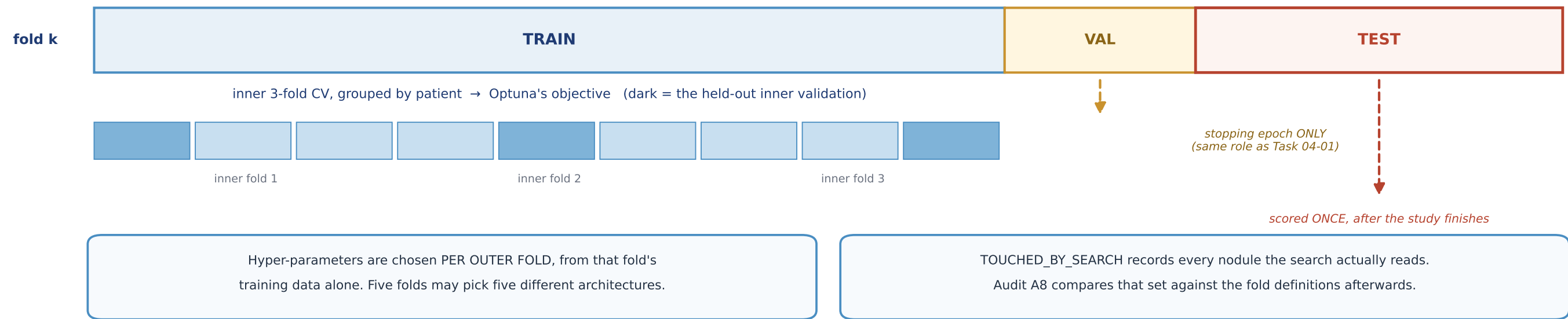


Freezing defines the search space, and caching makes the nested search affordable. Page 2 shows how the search is kept away from the test data.

Task 10 · Why the number is trustworthy, and what we report

Dr. Chattha's three constraints, each enforced by code and checked by an audit cell

4 Constraint (a) — Optuna never sees test or out-of-fold data

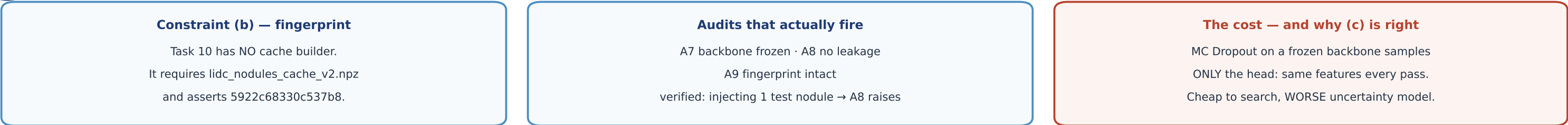


5 Constraint (c) — three rows, so the comparison means something



With only two rows, any difference confounds freezing with searching. The untuned probe is the pivot that separates them.

6 The guardrails, and the one thing freezing costs us



Structurally the same defect as the ViT drop_rate bug in Task 5 — which is exactly why the frozen variant stays an ablation row, not the baseline.